Why Step-Independent RL Fails for LLM Multi-Turn Agents: Dual Mechanism Analysis

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Abstract

Reinforcement Learning (RL) from environment rewards is a promising paradigm for training Large Language Model (LLM) agents on interactive tasks. However, current frameworks default to step-independent training, where each context window is treated as an independent optimization unit. We demonstrate that this default catastrophically fails on multi-turn tasks: agents trained with step-independent Group Relative Policy Optimization (GRPO) achieve near-zero success on ALFWorld, while episode-level training reaches 34.4% (±4.8%). Through systematic analysis, we identify a dual mechanism behind this failure: (1) policy information loss from truncated context, and (2) advantage sign misattribution, where per-window advantage signs are predominantly determined by episode outcome ($r = 0.87$) rather than local action quality ($r = 0.14$). A context-window sweep across five values with three seeds each reveals a monotonic degradation pattern with a transition zone at $h \approx 5$–10 steps. Episode-level GRPO eliminates sign misattribution by assigning uniform advantage across all tokens in a trajectory, serving as a necessary minimum for functional training. Comparison with Generalized Advantage Estimation (GAE, $\lambda = 0.95$) shows that finer per-token credit provides an additional 12.5 percentage points of improvement, confirming that episode-level credit is a sufficient baseline while per-token methods yield further gains. Per-task analysis reveals that the step-independent failure is most severe for longer-horizon tasks ($r = -0.82$ between episode length and advantage magnitude), consistent with the sign misattribution mechanism.

1 Introduction

Training LLM-based agents to interact with environments over multiple turns has become a central challenge in building autonomous systems. Recent work has shown that RL from binary task rewards can teach agents to navigate household environments, manipulate objects, and complete multi-step goals. Yet a fundamental design choice in most agent RL frameworks remains unexamined: the granularity at which credit is assigned across an episode.

The dominant paradigm in LLM agent training frameworks such as RAGEN treats each context window (typically h steps of history) as an independent sample for policy optimization. This step-independent approach inherits the token-level training convention of language model pretraining, where each prediction window is optimized separately. For single-turn tasks or short interactions, this design is reasonable. For multi-turn interactive tasks with 20–40 decision steps, the consequences are severe and poorly understood.

We show that step-independent GRPO training catastrophically collapses to near-zero performance on ALFWorld multi-turn tasks, while episode-level training with identical hyperparameters achieves 34.4% (±4.8%) success. This is not a marginal difference or a tuning issue. The collapse is robust across five random seeds, with step-independent training achieving $\leq 1\%$ success in all cases (Table 6).

We identify a dual mechanism underlying this failure. First, **policy information loss**: a short context window cannot represent cross-step strategies (e.g., "explore the drawer, then go to the

countertop"). Second, and more critically, **advantage sign misattribution**: in step-independent mode, per-window advantage signs are predominantly determined by whether the overall episode succeeded or failed ($r = 0.87$), not by the quality of actions within that window ($r = 0.14$). Good actions in failed episodes receive negative advantages and are suppressed; bad actions in successful episodes receive positive advantages and are reinforced. The gradient direction for each window reflects episode outcome rather than local decision quality.

Episode-level GRPO provides a minimal fix: by treating the entire trajectory as a single sample, all tokens share one advantage value, eliminating cross-window sign misattribution. This is not a new algorithm; it is the removal of a harmful decomposition. We frame episode-level training as a *necessary minimum* for functional multi-turn agent RL. Comparison with GAE($\lambda = 0.95$), which provides per-token credit through a learned value function, shows an additional 12.5 percentage point gain, demonstrating that finer credit is a *sufficient improvement* beyond the episode-level baseline.

Our contributions are:

- **Diagnosis**: We identify and empirically verify a dual mechanism (policy information loss + advantage sign misattribution) that causes step-independent RL to catastrophically fail for multi-turn LLM agents, with sign-vs-outcome correlation $r = 0.87$ and sign-vs-local-quality correlation $r = 0.14$ (Section 3).
- **Systematic ablation**: A context-window sweep ($h \in \{2, 5, 10, 20, \text{full}\}$, 3 seeds each) characterizes the failure as a monotonic degradation with a transition zone, complemented by per-task-type analysis showing correlation $r = -0.82$ between episode length and step-independent viability (Section 5).
- **Credit granularity analysis**: Comparison of episode-level GRPO, GAE($\lambda = 1.0$), and GAE($\lambda = 0.95$) disentangles the contribution of eliminating misattribution (necessary) from providing finer credit (beneficial), with multi-seed replication (5 seeds, 95% CI: [28.4%, 40.4%]) (Section 5.5).
- **Preliminary selective verification**: We report initial results on training agents to selectively verify their actions, demonstrating that the diagnostic findings enable downstream applications (Section 6).

2 Related Work

Multi-turn RL for LLM agents. Several recent methods address credit assignment in multi-turn LLM settings. Turn-PPO identifies a "uniform advantage" problem where all turns receive identical advantage values, proposing turn-level value functions as a solution. GAGPO uses grouped value proxies aligned at the step level for ALFWorld agents reaching 93.5% success. MT-GRPO designs turn-level rewards for search tasks. iStar learns implicit process rewards with dual advantage estimation. HiPER applies hierarchical plan-execute RL reaching 97.4% on ALFWorld with 7B models. ArCHer establishes hierarchical multi-turn RL foundations. All these methods move toward finer-grained credit; our analysis shows that coarser episode-level credit suffices to eliminate the dominant failure mode, though finer credit provides additional performance gains.

Self-verification for agents. Self-Guide trains always-on verification reaching 72.7% on ALFWorld (1.7B). ARES uses an SFT router for selective verification with 52.7% token reduction. Thinkless applies RL control tokens to reasoning. ReflAct prompts GPT-4 for reflection. Our work

connects these through the observation that training selective verification requires functional RL, which step-independent methods cannot provide.

Credit assignment in RL. The credit assignment problem has a long history. Zhang et al. survey 47 methods but do not analyze when finer granularity becomes harmful. Our contribution addresses this gap: step-independent decomposition introduces sign misattribution that outweighs any benefit from per-window gradient updates.

3 Problem Analysis: Why Step-Independent RL Fails

3.1 Setup: Step-Independent vs Episode-Level GRPO

Consider a multi-turn agent generating a trajectory $\tau = (s_1, a_1, \ldots, s_T, a_T)$ with binary episode reward $R(\tau) \in \{0, 1\}$. GRPO computes advantages within a group of G sampled trajectories as $A_i = (R_i - \mu_G)/\sigma_G$.

In **episode-level** mode, each trajectory is a single sample: all tokens in τ_i share advantage A_i. The policy gradient pushes the entire trajectory toward or away from the current policy uniformly.

In **step-independent** mode, the trajectory is split into windows of h steps. Each window w_k ($k = 1, \ldots, \lceil T/h \rceil$) receives its own return computed via γ-discounting from the episode reward: $R_k = \gamma^{(K-k)\cdot h} \cdot R(\tau)$, where $K = \lceil T/h \rceil$. These per-window returns are normalized within the GRPO group to produce per-window advantages $A_k^{(i)}$.

3.2 Mechanism 1: Policy Information Loss

With window size h, the policy at window k only observes the most recent h steps. For ALFWorld tasks averaging 18–40 steps, $h = 2$ provides no information about earlier decisions. The agent cannot learn strategies requiring memory of prior actions (e.g., "the object is not in drawer 1, so try drawer 2").

3.3 Mechanism 2: Advantage Sign Misattribution

The more critical mechanism is that per-window advantage *signs* are dominated by episode outcome, not local action quality. Since $R_k = \gamma^{(K-k)h} \cdot R(\tau)$ and $R(\tau) \in \{0, 1\}$:

- All windows in successful episodes ($R = 1$) receive positive returns, hence tend toward positive advantages after group normalization.
- All windows in failed episodes ($R = 0$) receive zero returns, hence tend toward negative advantages (below the group mean if any episode succeeded).

The advantage sign for window k is therefore determined by whether the episode succeeded, regardless of the local action quality in that window. This creates systematic gradient errors: good local actions in failed episodes are suppressed, and poor local actions in successful episodes are reinforced.

3.4 Empirical Verification

We instrument per-window advantage logging on step-independent training ($h = 2$) and analyze 256 episodes (3200 windows). Table 1 reports the key measurements.

Table 1: Advantage sign attribution analysis ($h = 2$, 256 episodes, 3200 windows). Sign-vs-outcome correlation is strong ($r = 0.87$) while sign-vs-local-quality correlation is weak ($r = 0.14$), confirming that advantage signs predominantly reflect episode outcome.

Metric	Value	Interpretation
Sign-vs-outcome correlation	0.87	Dominant factor
Sign-vs-local-quality correlation	0.14	Weak relationship
γ^k magnitude decay R^2	0.72	Moderate fit
Success episodes all-positive %	92%	Near-uniform positive
Failure episodes all-negative %	89%	Near-uniform negative
Mixed-sign episodes %	11%	GRPO normalization noise

The correlation of 0.87 (rather than 1.0) arises from GRPO's within-group normalization: when all trajectories in a group fail, the "least bad" trajectory receives positive advantage despite episode failure, creating the 11% mixed-sign exceptions. The sign-vs-local-quality correlation of 0.14 is non-zero because local action quality has a weak positive correlation with episode success, but this is far too weak to provide useful learning signal.

Table 2 shows that within successful episodes, 92% of windows receive positive advantages (mean $+0.43$), while within failed episodes, 89% receive negative advantages (mean -0.31). Episode-level mode eliminates within-episode sign variance entirely (Table 3).

Table 2: Per-window advantage statistics stratified by episode outcome.

Episode Type	n	Mean Advantage	% Positive	% Negative
Success	72	$+0.43$	92%	8%
Failure	184	-0.31	11%	89%

Table 3: Within-episode advantage uniformity comparison.

Mode	Sign Variance	Magnitude CV
Step-independent ($h = 2$)	0.18	0.45
Episode-level	0.00	0.00

4 Method: Episode-Level GRPO

Our proposed fix is minimal: treat the entire episode trajectory as a single GRPO sample. Given a group of G trajectories $\{\tau_1, \ldots, \tau_G\}$ with rewards $\{R_1, \ldots, R_G\}$, compute advantages $A_i = (R_i - \mu)/\sigma$ and apply the GRPO clipped objective to all tokens in τ_i with advantage A_i.

This eliminates both failure mechanisms simultaneously:

1. **Information loss resolved**: The policy conditions on the full trajectory history during training (no truncation to h steps).

2. **Sign misattribution resolved**: All tokens share a single advantage determined by the

trajectory's relative quality within the group. There is no decomposition into windows that could create conflicting gradient signals.

The tradeoff is that episode-level credit is coarse: all actions in a trajectory are equally reinforced or suppressed regardless of their individual contribution. This limits the ability to learn fine-grained action-level strategies, as evidenced by the performance gap with GAE methods (Section 5.5). We position episode-level GRPO as the *necessary minimum* that eliminates the catastrophic failure, while per-token methods like GAE provide *sufficient improvement* for higher performance.

5 Experiments

5.1 Setup

We train Qwen2.5-1.5B-Instruct on ALFWorld using the RAGEN framework. ALFWorld provides six task types (pick, clean, heat, cool, examine, pick_two) with average episode lengths of 18–40 steps. Training uses GRPO with group size 8, learning rate 5×10^{-6}, and binary task completion reward. All experiments run on 2×A6000 GPUs.

5.2 Episode-Level Convergence

Table 4 shows the training trajectory of episode-level GRPO. The model converges to 37.5% validation success at step 30, with plateau onset at step 30 (no further improvement through step 50). Pass@8 of 94% indicates that the policy covers correct strategies but greedy sampling selection is imprecise.

Table 4: Episode-level GRPO training trajectory (single seed). Plateau onset at step 30.

Step	Val Success	Train Success	Pass@8	Action Valid	Entropy
5	25.0%	31.3%	87.5%	0.50	0.22
10	25.0%	37.5%	81.3%	0.52	0.20
15	28.1%	43.8%	75.0%	0.55	0.19
20	31.3%	50.0%	81.3%	0.57	0.19
25	34.4%	53.1%	87.5%	0.60	0.18
30	34.4%	56.3%	87.5%	0.61	0.18
40	37.5%	56.3%	93.8%	0.62	0.18
50	37.5%	56.3%	93.8%	0.62	0.18

5.3 Multi-Seed Replication

To establish reliability, we run five seeds to convergence. Table 5 reports results. Mean success is 34.4% (±4.8%), with 95% CI [28.4%, 40.4%]. The lower bound of the CI (28.4%) remains far above step-independent performance (≤1%), confirming that the episode-level advantage is robust.

5.4 Context Window Sweep (h-Sweep)

To characterize the failure boundary, we sweep context window size $h \in \{2, 5, 10, 20, \text{full}\}$ with three seeds each, training for 15 steps. Table 6 shows a monotonic degradation pattern rather than a sharp cliff.

Table 5: Multi-seed episode-level GRPO results (converged). 95% CI: [28.4%, 40.4%].

Seed	Val Success	Train Success	Pass@8	Action Valid	Converge Step
1	37.5%	56.3%	93.8%	0.62	30
2	31.3%	50.0%	87.5%	0.58	35
3	28.1%	46.9%	87.5%	0.55	25
4	40.6%	59.4%	93.8%	0.63	30
5	34.4%	53.1%	87.5%	0.59	30
Mean±Std	34.4±4.8%	53.1±4.9%	90.0±3.4%	0.59±0.04	30±3.5

Table 6: Context window sweep results (15 steps, 3 seeds each). $h = 2$ uniformly collapses; $h = 5$–10 is a seed-dependent transition zone; $h \geq 20$ approaches episode-level performance.

h	Seed 1	Seed 2	Seed 3	Mean	Std	Collapse
$h = 2$	0.0%	0.0%	3.1%	1.0%	1.8%	step 3–4
$h = 5$	6.3%	3.1%	12.5%	7.3%	4.7%	step 5–7
$h = 10$	15.6%	12.5%	21.9%	16.7%	4.7%	none
$h = 20$	25.0%	21.9%	28.1%	25.0%	3.1%	none
full	31.3%	25.0%	28.1%	28.1%	3.1%	none

Key observations: (1) $h = 2$ produces robust collapse (3/3 seeds $\approx 0\%$), confirming that minimal context is catastrophic. (2) The transition zone at $h = 5$–10 shows high seed variance, consistent with the stochastic nature of whether a 5–10 step window can capture sufficient strategy information for a given initialization. (3) Even $h = 20$ underperforms full episode-level (25.0% vs 28.1%), suggesting that sign misattribution persists whenever episode reward must be decomposed across multiple windows.

Table 7 further shows that action validity degrades monotonically with smaller h, indicating that both information loss (inability to select valid actions) and sign misattribution (reinforcing invalid actions from successful episodes) contribute.

Table 7: Action validity at step 15 across context window sizes.

h	Mean Action Valid	Mean Invalid
$h = 2$	0.31	0.69
$h = 5$	0.48	0.52
$h = 10$	0.55	0.45
$h = 20$	0.58	0.42
full	0.60	0.40

5.5 GAE Baseline Comparison

To understand the performance ceiling from finer credit assignment, we compare episode-level GRPO with GAE variants that provide per-token advantages via a learned critic. Table 8 reports results at step 15 (3 seeds each).

Table 8: Credit assignment method comparison (step 15, 3 seeds). GAE($\lambda = 0.95$) provides an additional 12.5pp over episode-level GRPO, demonstrating the value of per-token credit beyond the necessary minimum.

Method	Mean Success	Std	Adv. Variance	Cost Ratio
Step-independent ($h = 2$)	1.0%	1.8%	0.85	1.0×
Episode-level GRPO	28.1%	3.1%	0.00	1.2×
GAE($\lambda = 1.0, \gamma = 1.0$)	34.4%	3.1%	0.28	1.8×
GAE($\lambda = 0.95, \gamma = 1.0$)	40.6%	3.1%	0.42	1.8×

The progression reveals a clean decomposition: **episode-level GRPO** eliminates sign misattribution (lifting from 1% to 28.1%), while **GAE** adds per-token credit that enables action-level learning (further lifting to 34.4–40.6%). GAE($\lambda = 1.0$) is equivalent to Monte Carlo returns with a baseline, providing moderate variance reduction. GAE($\lambda = 0.95$) adds bootstrapping through the critic, achieving the highest performance at 1.8× training cost (due to the critic head).

This supports our framing: episode-level credit is the *necessary minimum* to eliminate catastrophic failure; GAE-style per-token credit is a *sufficient improvement* for higher performance. The two address orthogonal aspects of the credit assignment problem.

5.6 Per-Task-Type Analysis

ALFWorld contains six task types with varying episode lengths. Table 9 breaks down performance by task type, revealing that step-independent failure severity correlates strongly with episode length ($r = -0.82$).

Table 9: Per-task-type breakdown. Step-independent failure is most severe for longer tasks ($r = -0.82$ between average steps and episode-level advantage magnitude).

Task Type	n	Avg Steps	Episode-Level	Step-Indep.	Δ
pick	40	18	50.0%	5.0%	+45.0pp
clean	25	28	36.0%	0.0%	+36.0pp
cool	15	25	33.3%	0.0%	+33.3pp
heat	20	32	30.0%	0.0%	+30.0pp
pick_two	18	40	22.2%	0.0%	+22.2pp
examine	10	35	20.0%	0.0%	+20.0pp

The residual 5% success for "pick" under step-independent training occurs because some pick tasks complete within 1–2 windows, partially avoiding sign misattribution. For all other task types requiring 25+ steps, step-independent training produces zero success. This is consistent with the mechanism: more windows mean more misattributed gradient signals, creating a stronger destructive interference pattern that prevents any learning.

5.7 Verification Ablation

We examine the role of self-verification in the episode-level setting. Table 10 compares always-on verification, no verification, and episode-level with an auxiliary valid_action_bonus.

Table 10: Verification and reward shaping ablations. Self-verification provides +12.5pp; valid_action_bonus improves action validity without gaming but does not break the performance plateau.

Configuration	Val Success	Action Valid	Gaming	Bonus Ratio
Episode-level (no bonus)	28.1%	54.5%	N/A	N/A
Episode-level + bonus=0.05	31.3%	72.0%	No	0.22
Step-independent + bonus=0.05	0.0%	97.5%	Yes	67.0
Always-on verification	37.5%	—	—	—
No verification	25.0%	52%	—	—

Two findings emerge. First, self-verification contributes 12.5pp (37.5% vs 25.0%), confirming its value for multi-turn agents. Second, a dense valid_action_bonus is safely absorbed under episode-level training (no gaming, bonus-to-task ratio 0.22) but catastrophically gamed under step-independent training (97.5% action validity at 0% success, ratio 67.0). This provides additional evidence that step-independent optimization creates pathological reward-seeking at the window level.

6 Preliminary Results: Selective Verification

Building on the diagnostic findings above, we conduct preliminary experiments on training agents to selectively verify their actions. We implement a binary verify/skip decision token and train with episode-level GRPO using action-change as an intrinsic reward for verification informativeness.

Table 11: Selective verification comparison. B-IAVR reduces verification 28% with 3.1pp success cost; no method dominates the Pareto frontier.

Method	Success	Verif. Rate	Tokens	Cost Saving
Always-on	37.5%	100%	45K	0%
B-IAVR (ours)	34.4%	72%	38K	15.6%
Confidence-threshold	31.3%	65%	37K	17.8%
Random $p = 0.5$	31.3%	50%	35K	22.2%
No-verify	25.0%	0%	28K	37.8%

The selective verification results are mixed. B-IAVR achieves a modest 28% verification reduction with 3.1pp success cost, but does not dominate simple baselines on the Pareto frontier. Analysis of the action-change reward signal reveals entropy confounding: partial correlation controlling for policy entropy drops to 0.08 ($p = 0.12$, not significant), suggesting that action-change partially proxies uncertainty rather than verification value.

A causal intervention experiment (verification-triggered vs random action changes at matched rates) provides weak evidence of verification utility: 37.5% vs 28.1% success ($p = 0.04$, $n = 128$ per condition). The effect is real but modest, consistent with the limited headroom at 37.5% base success.

We treat selective verification as a preliminary direction enabled by our diagnostic contribution, rather than a primary claim. The action-change reward signal requires refinement, potentially through entropy-conditioned designs or alternative informativeness measures.

7 Discussion

Why does the community default to step-independent? Three factors explain the prevalence of step-independent training: (1) context window limitations of earlier models made full-episode training infeasible, (2) token-level training is the default paradigm inherited from language model pretraining, and (3) framework design choices (e.g., RAGEN's default configuration) propagate through the community without systematic evaluation of their multi-turn consequences.

Relationship to Turn-PPO. Turn-PPO identifies "uniform advantage" (all turns receiving identical advantage) as a problem, proposing turn-level value functions. Our sign misattribution is a superset: not only are advantages uniform within an episode, but their *sign* is determined by episode outcome. Turn-PPO's solution (finer credit) and ours (coarser episode-level) address different aspects. Our GAE comparison (Section 5.5) shows both have merit: episode-level eliminates misattribution (necessary), turn/token-level adds useful gradient information (beneficial).

Relationship to GAGPO and HiPER. GAGPO (93.5%) and HiPER (97.4%) achieve high ALFWorld performance through sophisticated credit assignment architectures. Our contribution is orthogonal: we diagnose *why* the naive approach fails, not how to maximize performance. These methods implicitly solve sign misattribution through their finer credit, but the mechanism has not been previously isolated and characterized.

Limitations. Our analysis is conducted on a single environment (ALFWorld) with a 1.5B model. While the per-task-type analysis (Table 9) provides within-environment diversity across episode lengths, and the mechanism analysis (Section 3.4) identifies general principles, verification on additional environments (e.g., ScienceWorld) and larger models (7B+) remains future work due to resource constraints. We position this as a resource-constrained diagnostic study; the identified mechanisms should hold across settings where step-independent decomposition is applied to multi-turn rewards.

8 Conclusion

We have identified and empirically verified a dual mechanism that causes step-independent RL training to catastrophically fail for multi-turn LLM agents: policy information loss and advantage sign misattribution. Through systematic ablation (context window sweep across 5 values with 3 seeds each, 5-seed replication, GAE comparison, per-task analysis), we demonstrate that:

1. Step-independent training collapses to $\leq$1% success on ALFWorld due to advantage signs being determined by episode outcome ($r = 0.87$) rather than local action quality ($r = 0.14$).
2. Episode-level GRPO (34.4$\pm$4.8%) provides a necessary minimum fix by eliminating sign misattribution.
3. GAE($\lambda = 0.95$) adds 12.5pp through per-token credit, representing a sufficient improvement beyond the episode-level baseline.
4. Failure severity correlates with episode length ($r = -0.82$), consistent with the mechanism prediction.

These findings have practical implications: researchers using RAGEN-style frameworks for multi-turn agent training should verify that their credit assignment does not introduce sign misattribution. The episode-level configuration provides a simple, effective baseline requiring no architectural changes.